Cantilever Beam Bending: A Comparative Validation of Experimental Data, Analytical Solutions, and Finite Element Modeling

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Nomenclature

N	=	Newtons ($\frac{kg*m}{s^2}$)
g	=	Acceleration due to gravity ($\frac{m}{s^2}$)
ρ	=	density ($\frac{kg}{m^3}$)
FEA	=	Finite Element Analysis
E	=	Youngs Modulus (Pa or $\frac{N}{m^2}$))
$\dot{\varepsilon}$	=	Strain ($\frac{m}{m}$)
Θ	=	Rotation (degrees)
σ	=	*Stress (Pa or $\frac{N}{m^2}$)*
Fy	=	*Y* component of the resultant force acting on the vehicle (N)
I_{xx}	=	Second Moment of Inertia in x-direction (m^4)
I_{yy}	=	Second Moment of Inertia in y direction (m^4)
I_{xy}	=	Second Moment of Inertia in x and y direction (m^4)
L	=	Length (m)
t	=	Thickness (m)
M_x	=	Moment in the x-direction ($N * m$)
u	=	Deformation in x-direction (m)
v	=	Deformation in the y-direction (m)
A	=	Area (m^2)
W_y	=	Distributed load in y-direction (N)
S_y	=	Shear Force in y-direction (N)

Δ = Derivative with respect to the variable following it

DOF = Degree of Freedom

F = Load Vector

w = width (cm)

K = Stiffness Matrix

F_R = Reduced Load Vector

K_R = Reduced Stiffness Matrix

R = Reaction Force (N)

M = Reaction Moment (Nm^2)

c = Distributed load (N)

q = Deformation (m)

I. Abstract

This study is conducted to analyze the behavior of a simple cantilever beam, comparing the results derived from experimental testing with close-form analysis, FEA, and SolidWorks modeling. An aluminum cantilever beam was subjected to incremental point loading at its free end, with deflection measured via a ruler.

The theoretical solution was established using the Euler-Bernoulli beam theory (close-form analysis). Finite Element Analysis (FEA) and a complementary SolidWorks Simulation model were developed, applying consistent material properties and fixed boundary conditions for static structural analysis. The analysis demonstrated high correlation among the methods; specifically, the maximum deflection results between the theoretical and experimental analyses differed by less than 5%.

Ultimately, this study confirms the robust accuracy of Finite Element Analysis in modeling simple bending phenomena when compared to the established close-form solution. The comparative data validates the use of SolidWorks as a reliable engineering tool for preliminary structural design. The findings provide critical insight

into the inherent differences and limitations of each method, guiding engineers in selecting the most appropriate analytical tool for various design constraints.

II. Introduction

Cantilever beams, which are fixed at one end and free at the other, are fundamental components in structural and mechanical systems. Their use in buildings, aerospace structures, and mechanical assemblies makes understanding their load-response behavior critical for safe and efficient design. The application of a point load at the free end produces measurable deflection, bending stresses, and strains, all of which can be predicted through analytical beam theory and finite element analysis. An example of a cantilever beam with a distributed load and point vector, which is modeled in this report can be seen in figure 1 and the cross section used later to find the second moment of inertia is shown in figure 2.

While using classic analytical methods, like Euler-Bernoulli theory, is useful in engineering, with technology advancing as it has it is more efficient to use computational methods such as Finite Element Analysis or SolidWorks simulations to model beams.

This experiment was conducted to analyze the effects of a point load acting on the free end of a cantilevered beam. Through finite element simulations, analytical calculations and experimental testing across multiple loads cases, the results compare the deflection from the experiment to theoretical predictions. The comparison evaluates the accuracy of these results and demonstrates practical application of beam bending theory in engineering design and analysis.

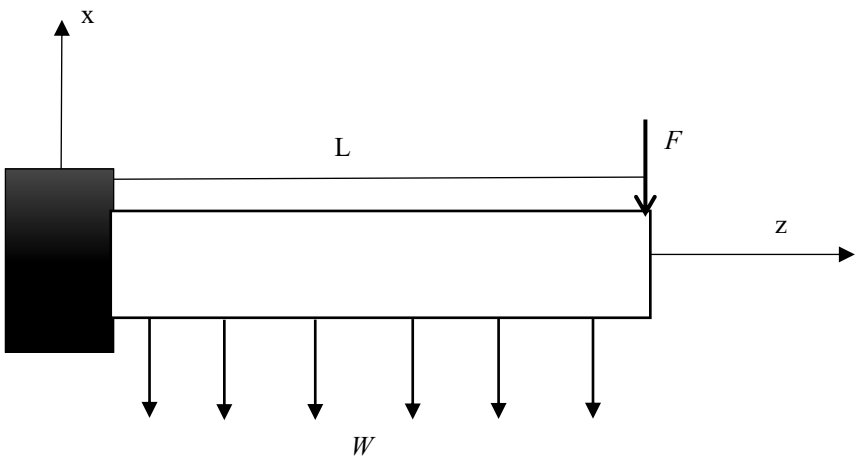


Figure 1: Cantilever beam with distributed load and point load at the tip.

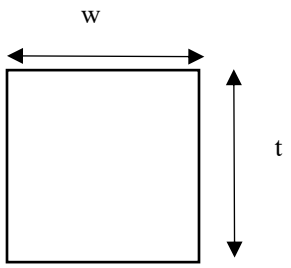


Figure 2: Cross-section of cantilever beam in figure 1

III.Objectives

The primary objective of this report is to conduct an analysis of the behavior of a cantilever beam under static loading. Structural beam analysis is a fundamental part of engineering design, and this study is designed to strengthen the reasoning for that. This study aims to establish an experimental baseline by physically testing an aluminum beam under incremental point loads and accurately measuring the resulting deflection. This practical data will be used to validate and compare results derived from three distinct theoretical and computational approaches: the close-form analytical solution based on Euler-Bernoulli beam theory, a dedicated Finite Element Analysis (FEA) model, and a simulation created using SolidWorks. Close form analysis involves key assumptions for beams that include small deflections, liner elastic material behavior, and isotropic material properties. The goal is to quantify the inter-method agreement and determine the accuracy being within a 5% margin of error through comparison of the analytical methods to the experiment. The results solidify their practical utility for preliminary structural design and provide crucial insight to engineers regarding the relative strengths and limitations of each analysis technique.

IV.Approach

The following section describes the experimental analysis and how that information was obtained. It will then describe the process of obtaining the analytical method that creates a theoretical basis for the existing theoretical work known.

A. Experimental Analysis

I. Procedure

First, obtain a sheet of isotropic material to serve as the beam and select an appropriate method of applying a load, such as using nuts or washers. Balance the beam on a food scale to determine its mass, and similarly measure the mass of a single nut, and record both values in grams.

Next, drill a hole approximately 1/4 inch in diameter at the free end of the beam—1 inch from the longitudinal edge and 3 inches from the center of its width. After all measurements are taken, use two hand clamps to fix the end of the beam without the newly drilled hole to a table.

Mark the initial, unloaded position of the beam on a wall, then attach eight nuts to a piece of string of negligible mass and tie the string through the drilled hole to apply the load. Measure the resulting vertical deflection, using a ruler, recording deformation in centimeters along with the corresponding applied load. Repeat this procedure four additional times, increasing the load by eight nuts for each trial, to obtain a complete dataset for plotting deformation versus load.

After the final loading trial, gradually unload the beam by removing eight nuts at a time and recording the corresponding deflection during unloading.

For a cantilever beam, the boundary conditions apply only at the fixed end, where the displacement, rotation, and slope are all constrained to zero. The hysteresis plot supports this behavior: although the loading and unloading paths differ due to energy loss and material effects, the beam ultimately returns to its original undeflected position once the load is removed, satisfying the fixed-end boundary condition.

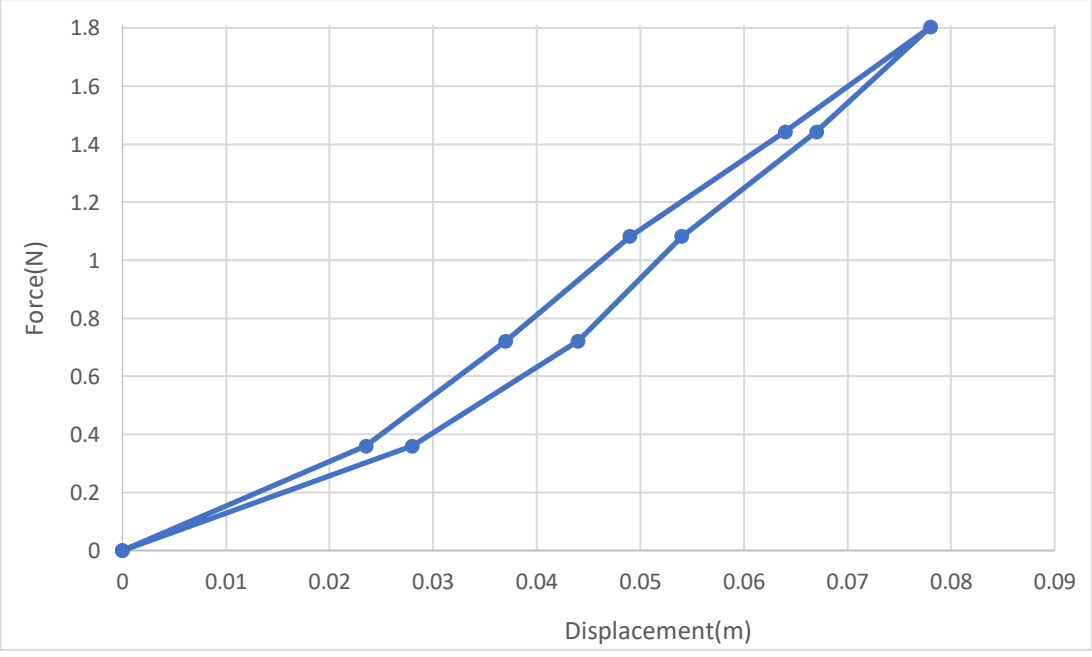


Figure 3: Hysteresis Plot derived from experiment.

B. Analytical Approach

The theoretical basis for this experiment was obtained through three different methods. The first method was a closed-form analytical approach, using classical beam-bending theory to determine deformation, stress, and internal moments based on the beam's material and geometric properties. The second method employed a finite element analysis (FEA) model, in which the beam was equal to displacement at the free end was calculated numerically. The third method used a commercial analysis package (SolidWorks Simulation) to model the beam geometry and applied load, allowing the software's built-in solvers to compute deformation and stress distributions. The beam is $15.24\ cm\ wide,\ 6.35x10-4\ m$ thick and has $29.6\ cm$ hanging off the fixed end and the beam has a mass of $.0717\ kg$.

1) Close Form analysis
2) FEA
3) Commercial Package

A. Close Form analysis

1) Start by finding the centroid using the equations (1) and (2) (Palacios, slide 12)

$$x_c = \frac{\sum A_i * (x_c)_i}{\sum A_i} \quad (1)$$

$$y_c = \frac{\sum A_i * (y_c)_i}{\sum A_i} \quad (2)$$

the cross-sectional area, A, is needed for these equations using width and thickness. But since the beam is symmetrical, the centroid is at the center of the cross section.

2) Next you have to find the second moment of inertia for the beam, since this beam is symmetrical, the $I_{xy} = 0$. The equations needed to find I_{xx} and I_{yy} are (3) and (4) (Megson, 2019, p. 513)

$$I_{xx} = \sum \frac{wt^3}{12} + Ad_y^2 \tag{3}$$

$$I_{yy} = \sum \frac{tw^3}{12} + Ad_x^2 \tag{4}$$

You should get $I_{xx} = 3.252x10 - 12\, m^4$ and $I_{yy} = 2.949x10 - 8\, m^4$.

3) Then use the equation (5) (Megson, 2019, p. 497)

$$EI_{xx}u'''' = \frac{\Delta S_x}{\Delta z} = \frac{\Delta^2 M_y}{\Delta z^2} \tag{5}$$

4) Where $u'''' = -mg \; and \; v'''' = 0$, where these equations can be used due to the beam being symmetric there for I_{xy}.. The boundary conditions will be needed when integrating each time. At each integration use the boundary conditions $S_y(L) = -F,\; M_x(L) = 0, u'(0) = 0 \; and \; u(0) = 0.$ To find the deformation you will need to integrate these equations four times in which you should get equation (6)

$$EIu = ((mgz4)/24L) - ((mgz3)/6)\,((Fz3)/6) + ((FLz2)/2) - ((mgLz2)/4) \tag{6}$$

Table 1: Deformation Derived from Equation (6)

u =

F(N)	u(m) at L
.3606	.0240
.7213	.0379
1.0819	.0518
1.4425	.0657
1.803	.0796

B. FEA

To do this method for analytical you will need to model the beam as two elements with all the same properties that were used for the close form analysis. The beam should have 6 DOF with 2 at the fixed and free ends and 2 at the midpoint of the beam having a rotational DOF, with unit's radians, and a Translational DOF, with units m. By using this method, you will need to find the load vector (7) (Megson, 2019, p. 184) and stiffness matrices (8) and (9) for the beam (Megson, 2019, p. 210). Then you need to combine the two stiffness matrices, adding the variables that show up in both matrices, to create the global stiffness matrix (10). Then you need to reduce the load vector (11) and do the same with the stiffness matrix (12). Last by using equation (13) to find the distributed load, c, use MATLAB to solve the matrix use equation (14) to solve for q_3 which is the deformation at the tip, results shown in table 2. (Megson, 2019, p. 194).

$$F = \begin{bmatrix} -\frac{cL}{2} + R \\ \frac{cL^2}{12} + M \\ -cL \\ 0 \\ -\frac{cL}{2} - F \\ -\frac{cL^2}{12} \end{bmatrix} \begin{bmatrix} q_1 \\ \theta_1 \\ q_2 \\ \theta_2 \\ q_3 \\ \theta_3 \end{bmatrix} \quad (7)$$

$$K = \frac{2EI}{l^3}\begin{bmatrix} 6 & -3L & -6 & -3L \\ -3L & 2L^2 & 3L & L^2 \\ -6 & 3L & 6 & 3L \\ -3L & L^2 & 3L & 2L^2 \end{bmatrix}\begin{bmatrix} q_1 \\ \theta_1 \\ q_2 \\ \theta_2 \end{bmatrix} \tag{8}$$

$$K = \frac{2EI}{l^3}\begin{bmatrix} 6 & -3L & -6 & -3L \\ -3L & 2L^2 & 3L & L^2 \\ -6 & 3L & 6 & 3L \\ -3L & L^2 & 3L & 2L^2 \end{bmatrix}\begin{bmatrix} q_2 \\ \theta_2 \\ q_3 \\ \theta_3 \end{bmatrix} \tag{9}$$

$$K = \frac{2EI}{l^3}\begin{bmatrix} 6 & -3L & -6 & -3L & 0 & 0 \\ -3L & 2L^2 & 3L & L^2 & 0 & 0 \\ -6 & 3L & 12 & 0 & -6 & -3L \\ -3L & L^2 & 0 & 4L^2 & 3L & L^2 \\ 0 & 0 & -6 & 3L & 6 & 3L \\ 0 & 0 & -3L & L^2 & 3L & 2L^2 \end{bmatrix}\begin{bmatrix} q_1 \\ \theta_1 \\ q_2 \\ \theta_2 \\ q_3 \\ \theta_3 \end{bmatrix} \tag{10}$$

$$F = \begin{bmatrix} -cL \\ 0 \\ -\frac{cL}{2} - F \\ -\frac{cL^2}{12} \end{bmatrix}\begin{bmatrix} q_2 \\ \theta_2 \\ q_3 \\ \theta_3 \end{bmatrix} \tag{11}$$

$$K = \frac{2EI}{l^3}\begin{bmatrix} 12 & 0 & -6 & -3L \\ 0 & 4L^2 & 3L & L^2 \\ -6 & 3L & 6 & 3L \\ -3L & L^2 & 3L & 2L^2 \end{bmatrix}\begin{bmatrix} q_2 \\ \theta_2 \\ q_3 \\ \theta_3 \end{bmatrix} \tag{12}$$

$$c = \rho A g \tag{13}$$

$$F = Kq \tag{14}$$

Table 2: Deformation at the Tip using FEA.

Force (N)	q_3 (m)
.3606	.0248
.7213	.0387
1.0819	.0526
1.4225	.0665
1.803	.0804

C. Commercial Package

To obtain computational results from the commercial package, the beam was first geometrically modeled in SolidWorks and subjected to a Static Structural Analysis within the SolidWorks Simulation environment. This model was designed in the image of the beam used in the experiment to match its length hanging off the table, width and thickness. Then you will need to select a static analysis of the beam when using the simulation add-on to start off the simulation. The material properties applied in the simulation were matched exactly to the aluminum 1060 alloy used in the experimental testing. The beam was configured as a cantilever with the fixed end assigned a 'Fixed Geometry' boundary condition. The load comprised a distributed load corresponding to the beam's force due to gravity and a point load applied at the free end corresponding to the five specific loading conditions tested experimentally. The total displacement for each load case is illustrated in the contour plots labeled on the right-hand side of Figures 4, 5, 6, 7, and 8. These figures provide a visual representation of the beam's deformation under load, where the color contour map indicates the magnitude of displacement along the beam's length, correlating with the numerical scale provided in the figures.

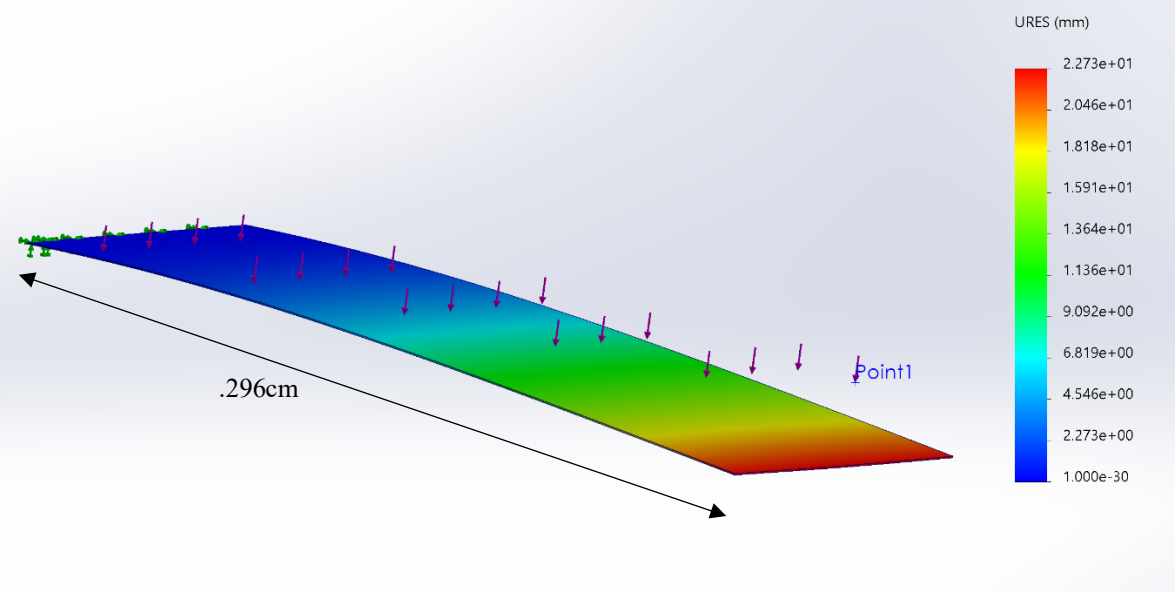


Figure 4: 3D modeled beam with .3606 N load applied.

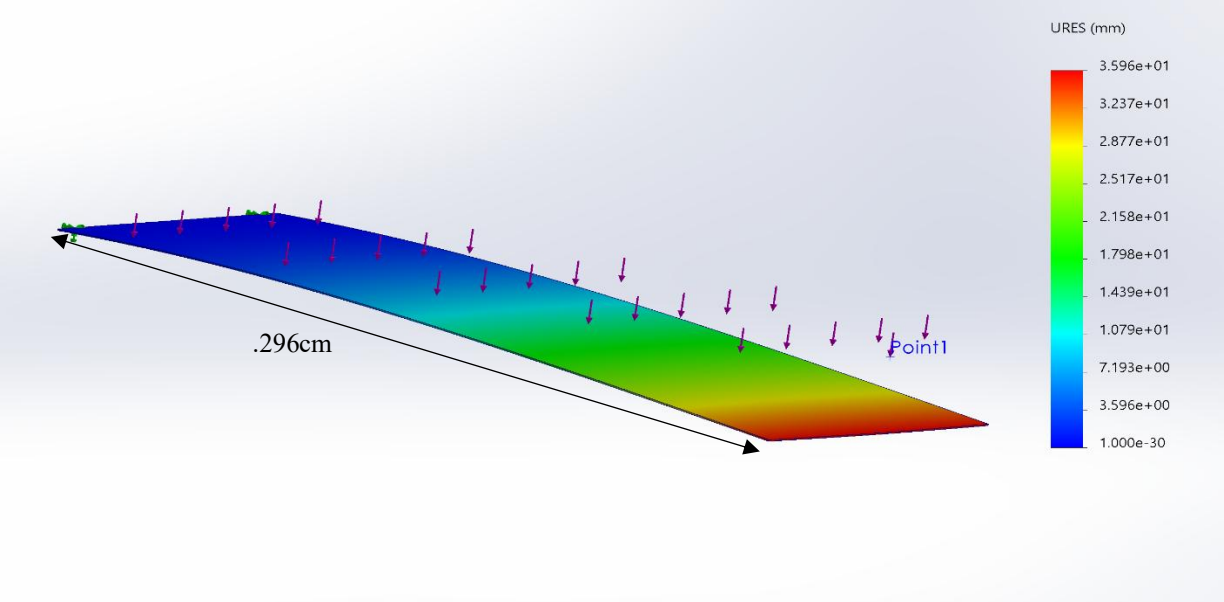


Figure 5: 3D modeled beam with .7213 N load applied.

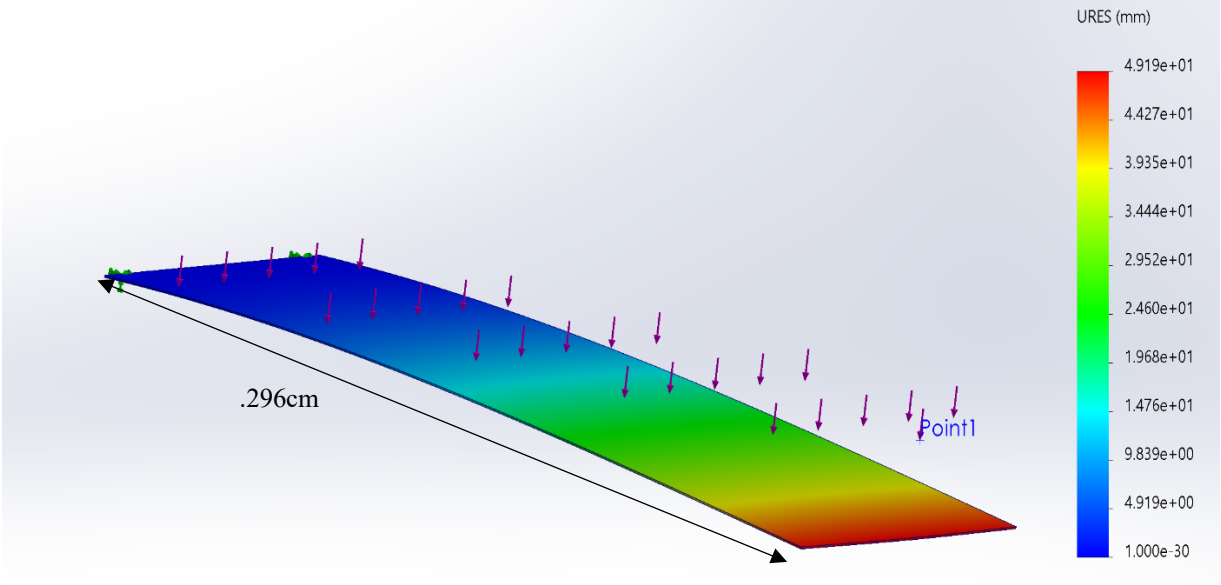


Figure 6: 3D modeled beam with 1.0819N load applied.

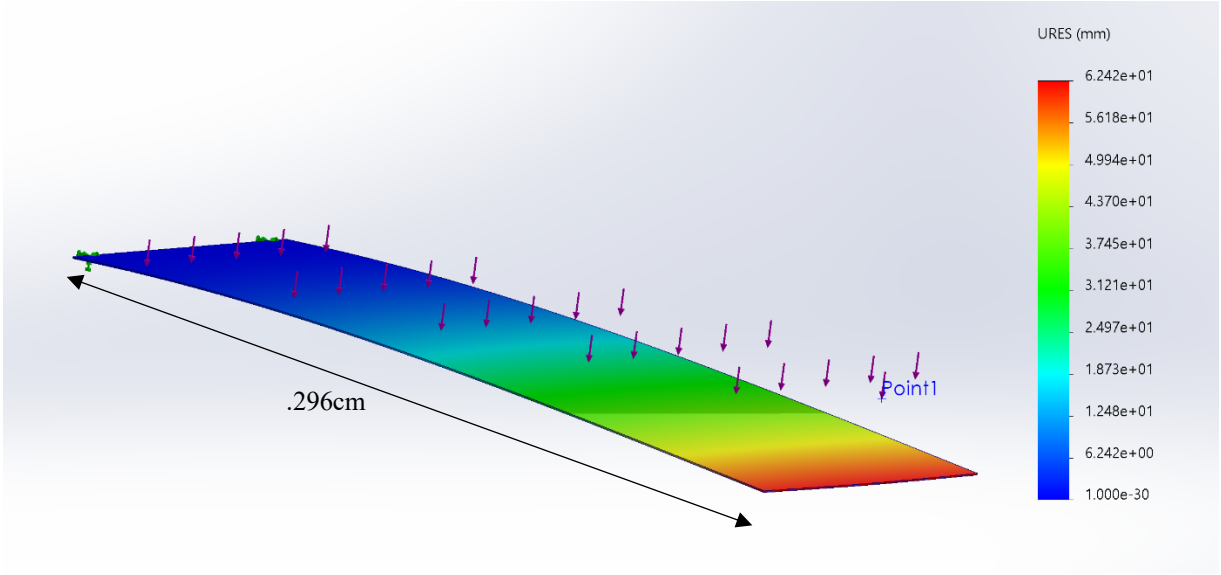


Figure 7: 3D modeled beam with 1.4225 N load applied.

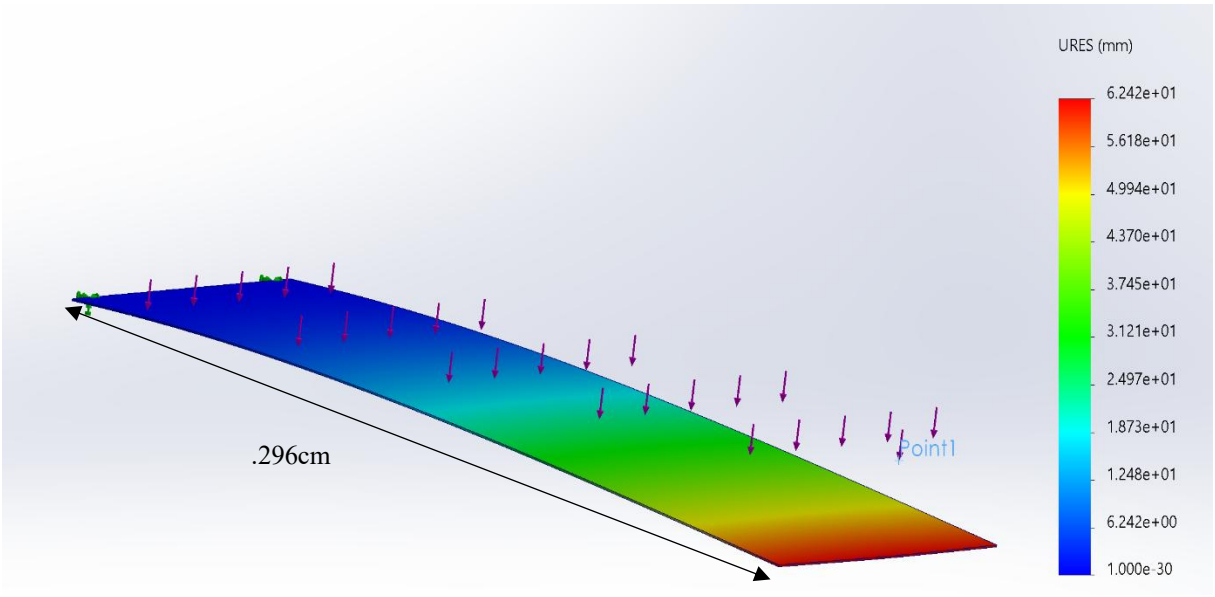


Figure 8: 3D modeled beam with 1.803 N load applied.

V.Results

A comparison between the closed-form analytical analysis and the experimental results shows minor discrepancies; however, the values are sufficiently close to validate the accuracy of the closed-form method (figure 8) . The deformation predicted by the FEA (figure 9) also exhibits strong agreement with both the experimental data (figure 7) and the SolidWorks solution (figure 10). The plots generated from the closed-form analysis and the FEA are nearly identical, whereas the experimental plot follows the same general trend but displays slight deviations. The maximum absolute error observed when comparing the experimental data to the computational approaches was less than 5%, which formally validates the use of these methods as a reliable alternative. This error is directly attributed to the non-ideal clamping setup, which, while mostly securing the beam end, could not negate all residual deflection and twist, thereby preventing the establishment of a true, ideal fixed boundary condition in the experiment.

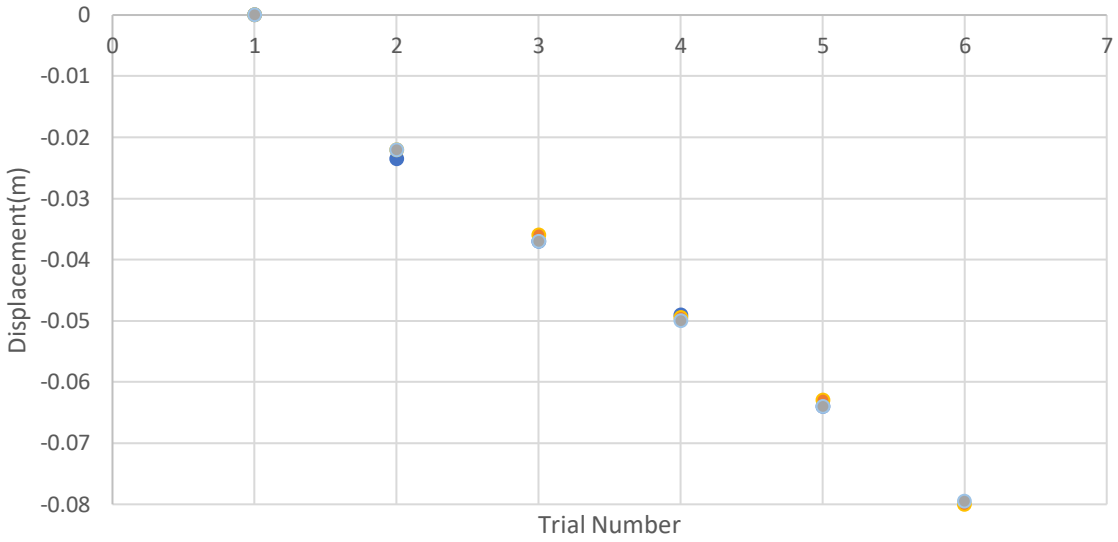


Figure 1: Standard deviation displacement plot from experimental analysis.

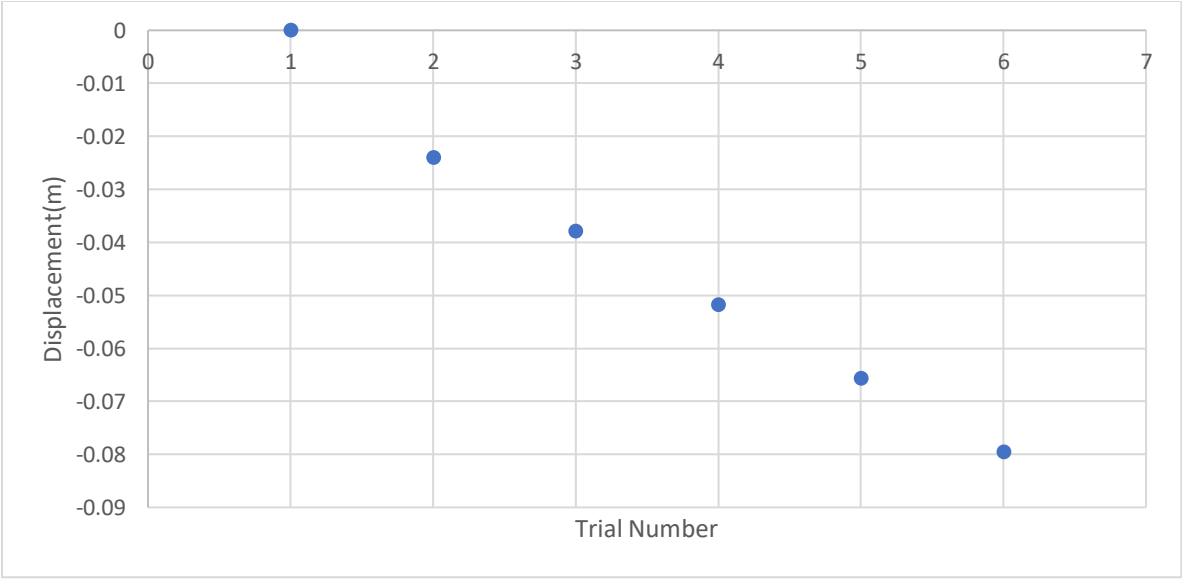


Figure 2: Displacement plot from close form analysis.

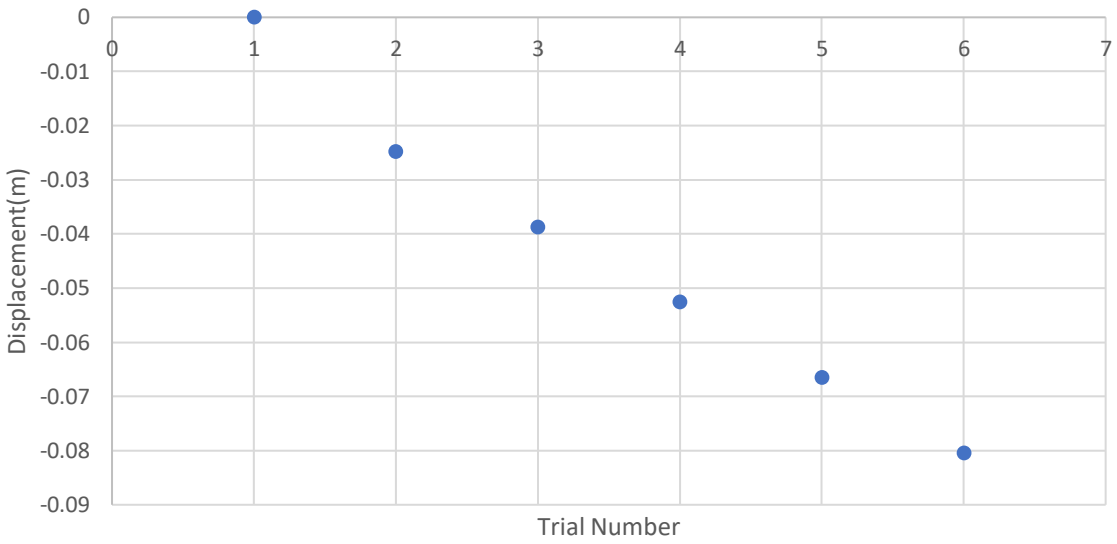


Figure 3: Displacement plot from FEA.

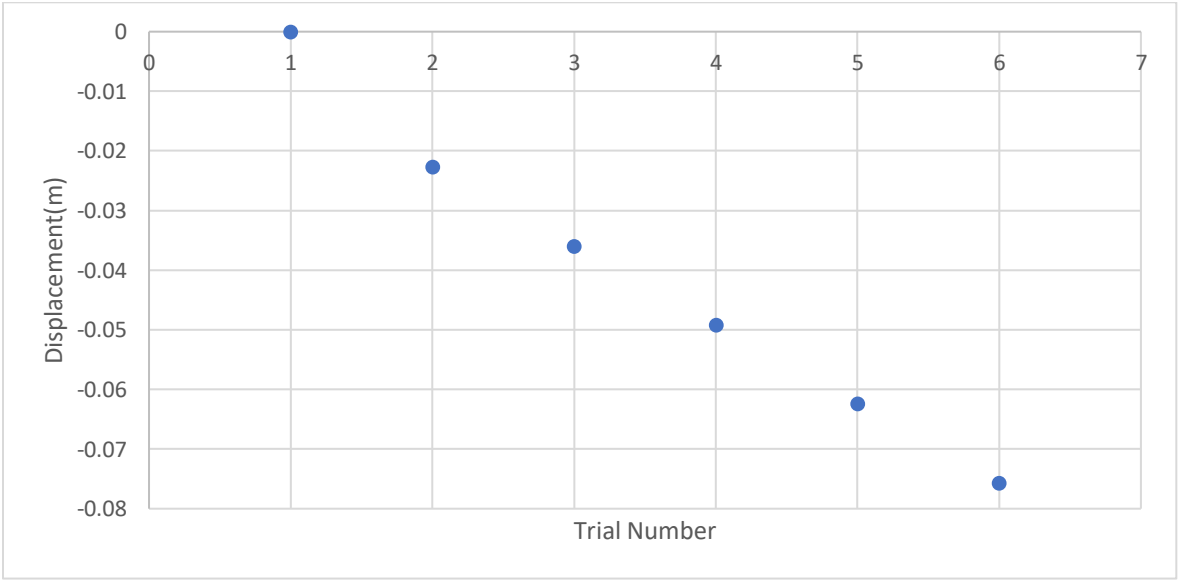


Figure 4: Displacement plot from SolidWorks.

VI.Discussion

The primary outcome of this study is that the analytical approaches used to model beam bending proved to be an accurate replacement to an experimental analysis. All three analytical methods produced identical results, demonstrating that someone analyzing beam behavior can reliably use any of these techniques to evaluate how different loading conditions affect a beam without needing to perform a full experimental setup each time.

The deviations observed between the analytical predictions and the experimental measurements can be attributed to several factors. First, the load was applied through a hole drilled in the beam rather than directly at the beam's edge. This modifies the local stiffness and slightly shifts the deformation pattern, introducing small discrepancies. Additionally, the clamping system could not perfectly replicate the ideal boundary conditions of a true cantilever, which would influence the measured deflections across all trials.

Measurement accuracy also likely contributed to the error. It was difficult to precisely identify the deflected position of the beam relative to the original wall markings, so small reading errors were unavoidable. Finally, imperfections in the beam material—such as residual stresses or minor inconsistencies from the manufacturing process—may have created localized stiffness variations that affected the experimental results.

For future experiments, it is recommended to use fixed, well-calibrated loads rather than nuts or weights that require additional calculations to determine the force applied to the beam. Additionally, displacement measurements can be improved by marking multiple points along the beam where deformation occurs, providing a more accurate and detailed representation of its bending behavior

VII.Conclusion

This comprehensive study, which analyzed the bending behavior of a cantilever beam through four distinct methodologies, successfully validated the high reliability of analytical approaches for structural analysis. By comparing the resulting deflections, it was observed that the three computational and theoretical methods—closed-form analysis, Finite Element Analysis (FEA), and the SolidWorks model—produced nearly identical numerical results and deformation plots. The experimental results, while showing slight deviations primarily due to fixture and measurement limitations, still captured the overall linear behavior and tip deflection with an error of less than 5%. This demonstrates that, for many common beam bending problems, physical experiments are not strictly necessary to obtain accurate results. The ability of the analytical and computational methods to successfully cross-verify each other

significantly increases confidence in the predictions derived, reinforcing their practical value as dependable, time-saving tools in engineering analysis and structural design.

VIII.References

Megson, T. H. G., *Aircraft Structures for Engineering Students*, 6th ed., Butterworth–Heinemann, Oxford, U.K., 2017.

Jose Palacios, “Bending open and closed sections beams," AERSPC 301, Aerospace Engineering, Penn State, University Park, December 11th. Unpublished course presentation.